13.4 Hot In-Place Recylcing

Hot In-place Recycling (HIPR) is an on-site, in-place method for rehabilitating pavements. The process is used to correct surface distress such as surface cracking, raveling, low friction values, and distortion confined to the wearing course such as corrugations and shoving. It cannot be used to correct structural failures. By eliminating the cost of transporting, processing, and stockpiling the RAP and new HMA, HIPR can be very economical compared with alternative treatments.

The process softens the existing HMA pavement structure with heat, removes or scarifies the surfacing material, mixes the material with a soft asphalt or a recycling agent and additional aggregate or HMA (if required), and places and compacts the new recycled asphalt surface on the roadway. One advantage of this process is that the thickness of any new overlay pavement can be reduced because the surface distress in the existing pavement is eliminated. Thus, current elevations and the overhead clearances are preserved.

The three categories of HIPR are surface recycling, repaving, and remixing. The first two processes involve heating, scarification, rejuvenation, and replacement of the top 1–1½ inches (25–37.5 millimeters) of the existing pavement. The repaving process combines surface recycling with the placement of a new HMA lift directly on the recycled but yet uncompacted material. The new material is not mixed with the recycled material but is compacted as a single layer. The recycled depth in the repaving process is still 1–1½ inches (25–37.5 millimeters) and the new overlay is placed up to a depth of 2 inches (50 millimeters). The remixing process involves blending virgin HMA directly with the scarified existing material in a pugmill and placing it to a depth of 2 inches (50 millimeters).

13.4.1 Pavement Assessment

HIPR is used to correct surface defects in an HMA pavement surface. It is not the appropriate treatment for a pavement that has base-course or subgrade failures. The thickness of the existing HMA layer should be at least 3 inches (75 millimeters) and should not contain a geotextile interlayer within the recycling depth. The presence of surface treatments, rubberized crack sealants, thermoplastic paint stripes, and specialty mixes (such as an open-graded friction course or an SMA mix containing an asphalt-rubber binder), may present problems and require approaches to surface rehabilitation other than HIPR.

13.4.2 Material Evaluation

The mix design of an HIPR project is developed using samples obtained from the existing roadway. The frequency of the sampling program will vary with the size and uniformity of the project. Samples should be obtained at a minimum of two sampling locations per lane-mile (or about one per lane-kilometer). The amount of material obtained from each sampling location will vary according to the amount of laboratory testing to be done. The sample for a mix design test should be larger than that needed when only thickness, gradation, and asphalt content are needed. Both core samples and chunk samples (obtained without water) should be taken. Large core samples are preferred (greater than 6 inches [150 millimeters] diameter). The samples should be tested to determine the in situ moisture content.

The core samples should be evaluated to determine the thicknesses of the various layers, the presence of paving geotextiles, and evidence of stripping or anything that may lead to delamination. After inspection, the cores should be trimmed to the depth of the recycled mix and the following properties should be evaluated:

- Bulk specific gravity
- Asphalt binder content
- Properties of the aggregate—gradation, coarse-aggregate angularity, fine-aggregate angularity
- Recovered binder properties determined using the PG binder testing procedures
- The maximum theoretical specific gravity of the mix, G_{mm}
- The volumetric properties of the mix—Va, VMA, and VFA

Binder Evaluation and Blending Charts

The first step in developing a mix design for an HIPR mix is to determine how to rejuvenate the asphalt binder in the existing pavement. As the existing pavement has been exposed to the elements, the asphalt binder in the surface mix has been exposed to air, water, and sunlight, resulting in the oxidation and hardening of the asphalt binder. One of the benefits of the HIPR process is the ability to rejuvenate this binder and return it to its original properties. This can be accomplished by adding a recycling agent, a soft asphalt, or a combination of the two.

The effect of adding a recycling agent or soft asphalt can be determined by measuring the physical properties of the recovered asphalt binder. The asphalt binder is tested to determine the penetration, viscosity, or $G^*/\sin\delta$ stiffness using a dynamic shear rheometer. This information is used with a blending chart. Figure 13.11 is a blending chart that uses PG binder criteria. The stiffness ($G^*/\sin\delta$) is determined of the recovered asphalt binder. This is plotted on the left-hand axis. The properties of the soft asphalt or rejuvenating agent are plotted on the right-hand axis. A line is drawn between the two points. Where that line crosses, the specification limit of one kilopascal is the percentage of new binder needed. This analysis

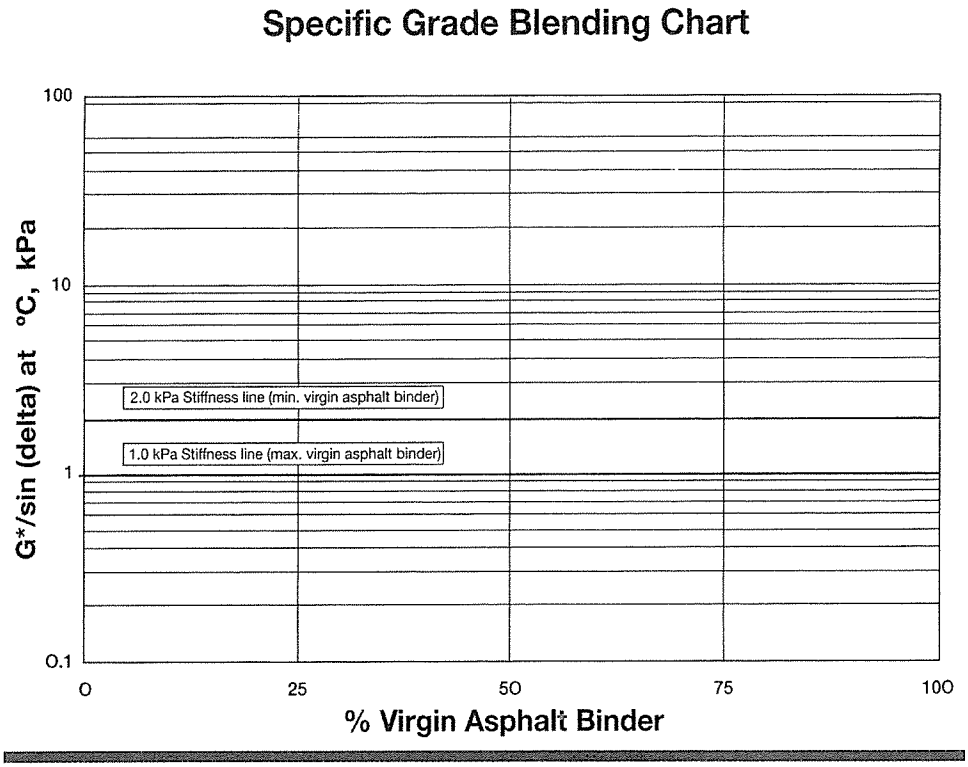


Figure 13.11 Blending Chart

should be verified by blending various percentages of the recycling agent or soft binder with the recovered binder and then testing the result to determine the changes in the physical properties that result from adding the recycling agent/new binder.

Mix Design

When the amount of asphalt, new aggregates, and new HMA has been determined from the blending processes previously described, they should be combined and compacted using appropriate mix design procedures for the traffic volume of roadway for which the mix is being designed. The volumetrics of the mix (air-voids, voids mineral aggregate, and voids filled with asphalt) should be determined and the properties of the combined asphalt binder should be checked to ensure that the needs of the project are being met. If these properties do not meet established criteria, the mix proportions should be adjusted.

13.4.3 HIPR Construction Processes

Surface Recycling

Surface recycling, or heater-scarification, consists of heating the top layer of an existing pavement, scarifying the softened HMA, adding a recycling

agent (if required), mixing the loose mix, spreading the mix with a floating screed, and compacting the recycled mix with standard HMA rollers. No virgin aggregate is added in this process. A wearing course of a surface treatment or a thin layer of HMA may be added on top of the surface recycled layer.

The drying and heating of the existing pavement surface is accomplished by one or more preheating units (see Figure 13.12). These units use indirect radiant or infrared heaters generally fueled with propane. The newest units use high velocity, super heated air that passes through small holes in a plenum passing near the pavement surface. This super-heated air is then captured and circulated through the heating unit to re-heat the air. Multiple units operate close to one another to maximize the penetration of the heat into the pavement. The preheating units should increase the temperature of the pavement without hot spots or scorching the existing pavement. The temperature of the scarified layer should be at least 230°F (110°C) and less than 300°F (150°C). The effectiveness of the preheating process depends on the temperature of the heat source, the ambient conditions (air and pavement temperatures), moisture content, and the length of time the pavement surface is exposed to the heat. If the ambient conditions are such that the pavement is not being properly heated, the time the surface is exposed to the heat source must be increased. This can be done by slowing the rate at which the source moves across the surface or by adding heating units.

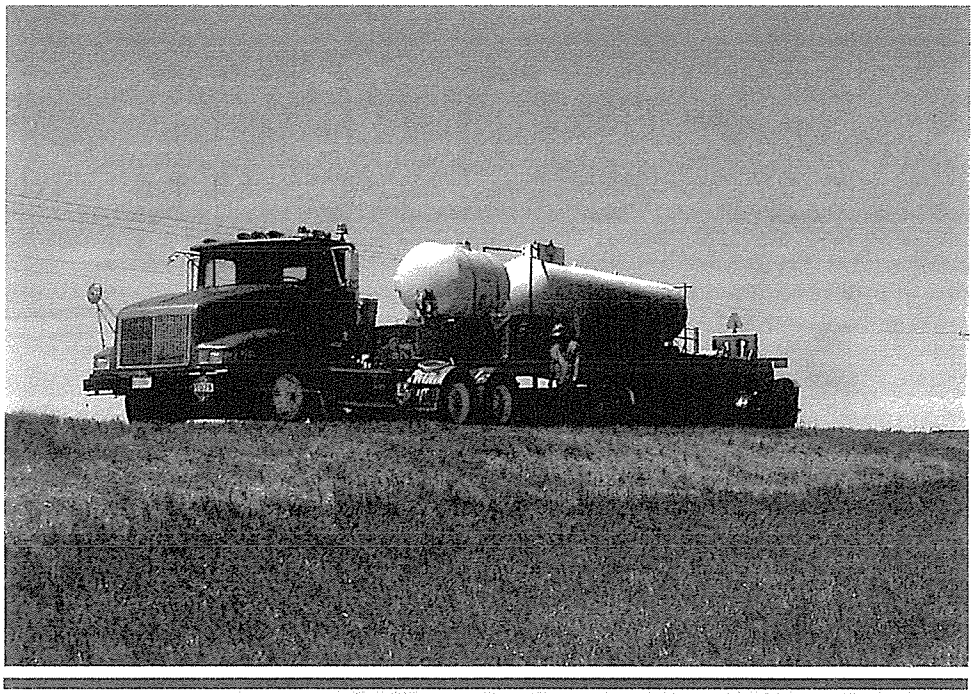


Figure 13.12 Surface Recycling Pre-heaters

After the road surface has been preheated, a heating/scarification unit passes over the surface (see Figure 13.13 for a close-up of the scarifying tines). If necessary, a recycling agent is added to the scarified layer. The control system for adding the recycling agent is linked to the forward speed of the scarifying unit. As previously discussed, the recycling agent rate will vary according to the properties of the asphalt binder in the HMA being scarified. Generally, it is applied at a rate of up to 0.5 gallon/yard^2 (2 litre/meter^2). The scarified material is struck off with a heated, vibrating screed using manual grade control.

Compaction is generally accomplished by using a pneumatic roller as a breakdown roller, followed by a static steel-wheeled or double-drum steel vibratory roller for finish rolling. Static rolling is preferred where very thin lifts are being compacted to avoid damaging the mixture.

The compacted recycled surface can be left as the wearing course on a low-volume roadway, or it can function as a pretreatment for an HMA overlay or surface treatment. Lower costs and higher production generally result if the placement of the wearing course and the scarifying operation are separate and independent operations that do not delay each other.

Repaving

The repaving process consists of heating, scarifying the softened pavement, applying a recycling agent to the scarified material, and spreading the scarified material to form a leveling course similar to the surface

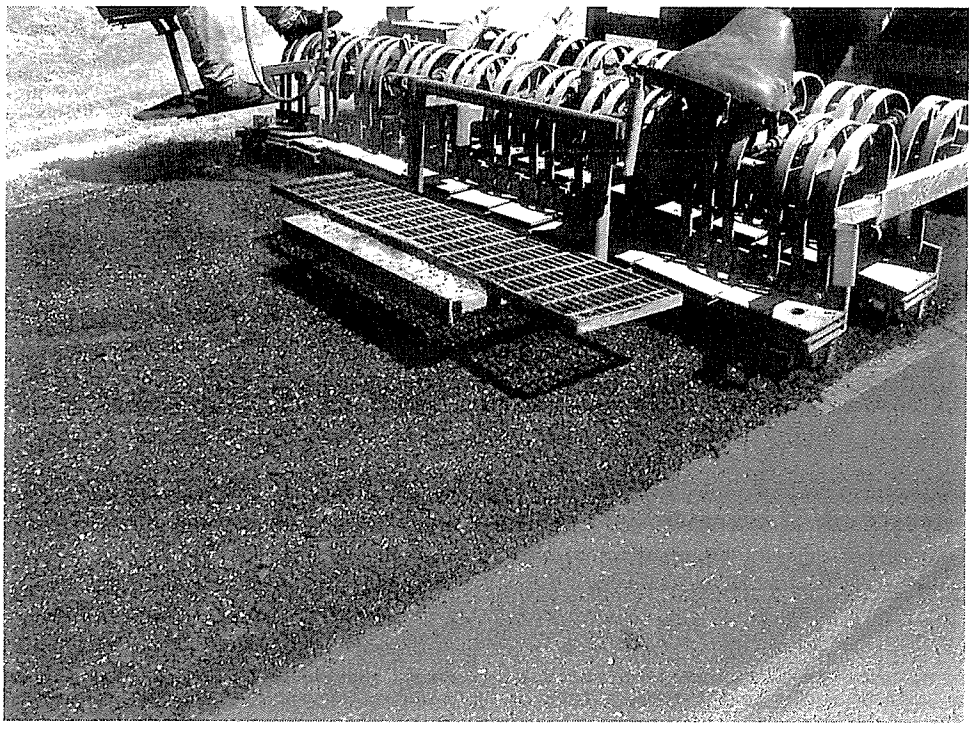


Figure 13.13 Close up of Scarifying Tines

recycling process, then placing a lift of new HMA directly on the recycled material before compaction. Figure 13.14 shows the new HMA being delivered to the repaver and deposited in the front receiving hopper. There it is conveyed the length of the equipment by a drag-slat conveyor to the paving screed, where it is placed, as in a routine HMA paving operation. Alternately, a coordinated two-stage process can be used where a conventional paving machine applies new HMA directly over the freshly heater-scarified pavement. Figure 13.15 shows the mixing and scarifying of the existing HMA surface prior to the new HMA being added to the roadway.

The combined thickness of the two layers is typically about 3 inches (75 millimeters); a combined thickness greater than this can result in placement and compaction problems. After placement of the mix, compaction is accomplished with a conventional roller train using the compaction procedures discussed in Chapter 9, "Hot Mix Asphalt Compaction."

Remixing

Remixing consists of heating the roadway to a depth of 1½–2 inches (37.5–50 millimeters); scarification and collection of the softened material into a windrow; mixing the material with virgin aggregates, a recycling agent, and/or new HMA, if required, in a pugmill; and laying the recycled mix in a single homogeneous HMA layer. This method is used when the repaving procedure is not sufficient to make the necessary corrections to an existing roadway. It can be used to make changes in aggregate gradation, improve skid resistance, and adjust the mix to improve rutting resistance.

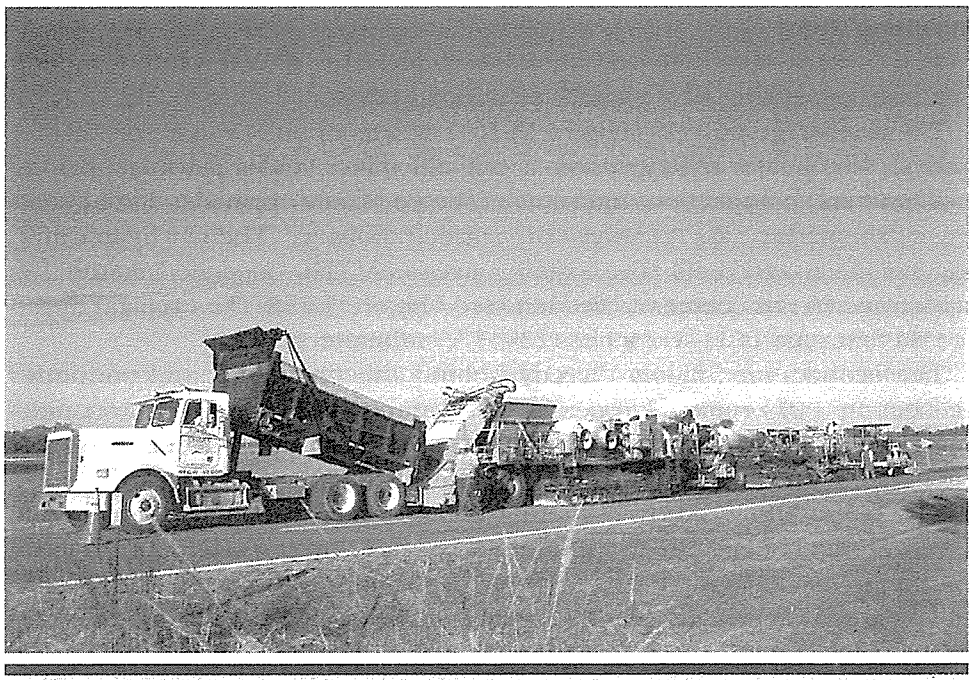


Figure 13.14 Repaver

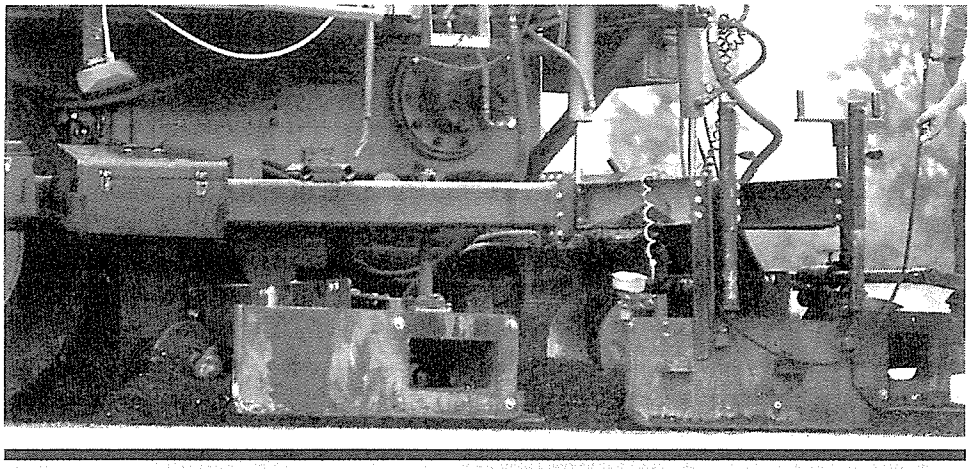


Figure 13.15 Repaver Mixer

Remixing can be either a single-stage or a multistage process (see Figure 13.16). The treatment depth for the single-stage process is typically 1–2 inches (25–50 millimeters). The treatment depth of the multistage method is 1½–3 inches (37.5–75 millimeters).

The heating of the existing pavement is done in much the same way as the surface recycling procedure. Scarifiers and/or hot milling drums (Figure 13.18) loosen the softened material, which is then augered to the center of the unit. These units have a moldboard for collecting the RAP. Depth control is accomplished by automatic sensors operating off a short reference ski or matching shoe.

In the single-stage method, the loose material is directed into a mixing chamber consisting of a pugmill or a mixing drum.

The multistage process (Figure 13.16) consists of one or more pre-heater unit(s), two heater-milling units, a pugmill mixer, and a standard paving machine and compaction equipment. The pre-heater provides initial heating of the surface and removes the surface moisture. Then, a heater-milling unit heats and mills the existing pavement. The loosened material is windrowed in the center of the lane (see Figure 13.18). The typical depth for this first pass is 1–1½ inches (25–37.5 millimeters).

The second unit follows directly behind the first unit and continues the heating and removal process. The distance between these two units must be kept to a minimum to ensure the least amount of heat loss. The second unit picks up the milled material from the first unit as it mills 1-inch below the windrow. This unit lifts and removes the loose material. If required, new HMA is dropped into a hopper on the second unit. From the hopper, it is dispensed evenly over the windrow from the first unit. This material is carried over the heating chamber by a slat conveyor that has a variable-speed control that matches the speed of the second unit. It is dropped into the windrow created by the final milling head. This milling head mills the second 1-inch layer of the softened material into a windrow.

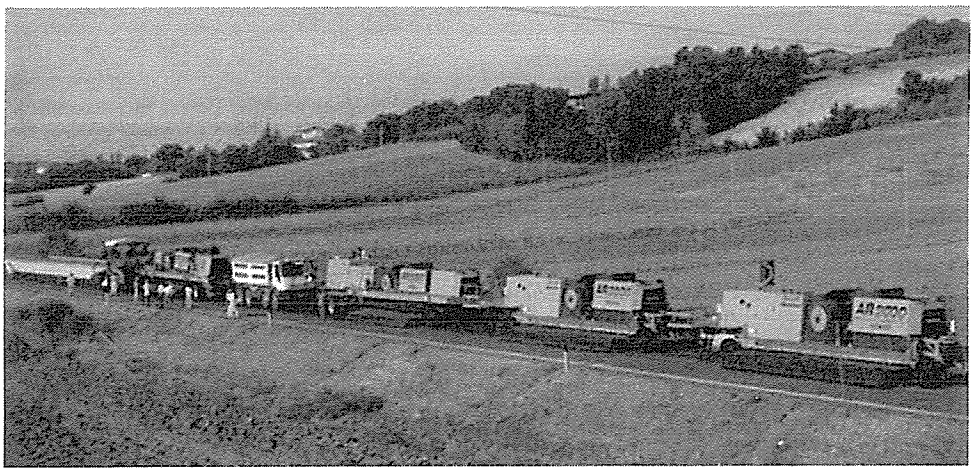


Figure 13.16 Multistage Remixing Train

All the materials are combined in a continuous pugmill unit to produce a homogeneous mix—of the material from the roadway and new HMA.

This material is laid on the roadway with a standard laydown machine. Breakdown rolling, typically done with a heavy pneumatic roller, should follow closely behind the laydown machine. The recycled mix cools quickly and requires both breakdown and finishing rollers to complete the compaction process quickly. Finish rolling is accomplished with a double-drum vibratory roller.

13.4.4 Inspection

The following details require attention:

- **Density.** Density requirements should be the same as for new HMA.
- **Mixture requirements.** Samples should be taken from the roadway and compacted in the lab to ensure that the mixture being placed meets the design volumetric requirements.
- **Uniform placement.** The material should be placed in a uniform manner that will depend on the nature of the mixture being recycled and on the uniformity of the feed systems. The mix should be placed so that no segregation occurs in the pavement and so that the smoothness requirements are met.
- **Depth control.** This is critical if a quality produce is to be obtained.
- **Vegetation damage control.** Vegetation along the roadway can be damaged by the heat used during the process. Spraying the vegetation with water can help minimize this problem.
- **Traffic control maintenance.** Typically, traffic can use the pavement within one hour after compaction has been completed.

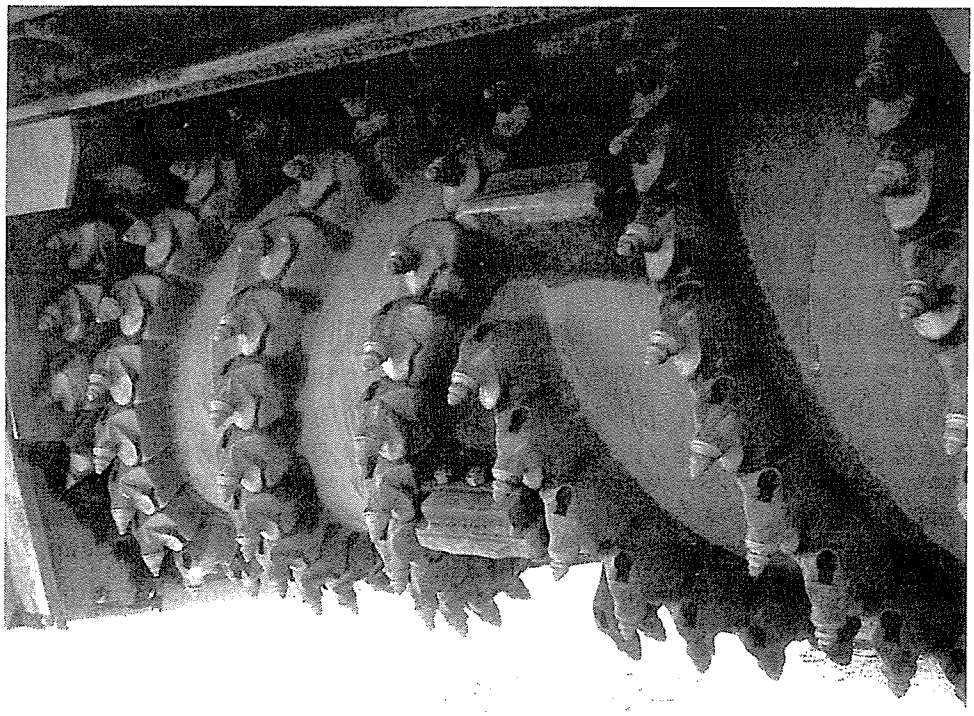


Figure 13.17 Milling Head on First Unit

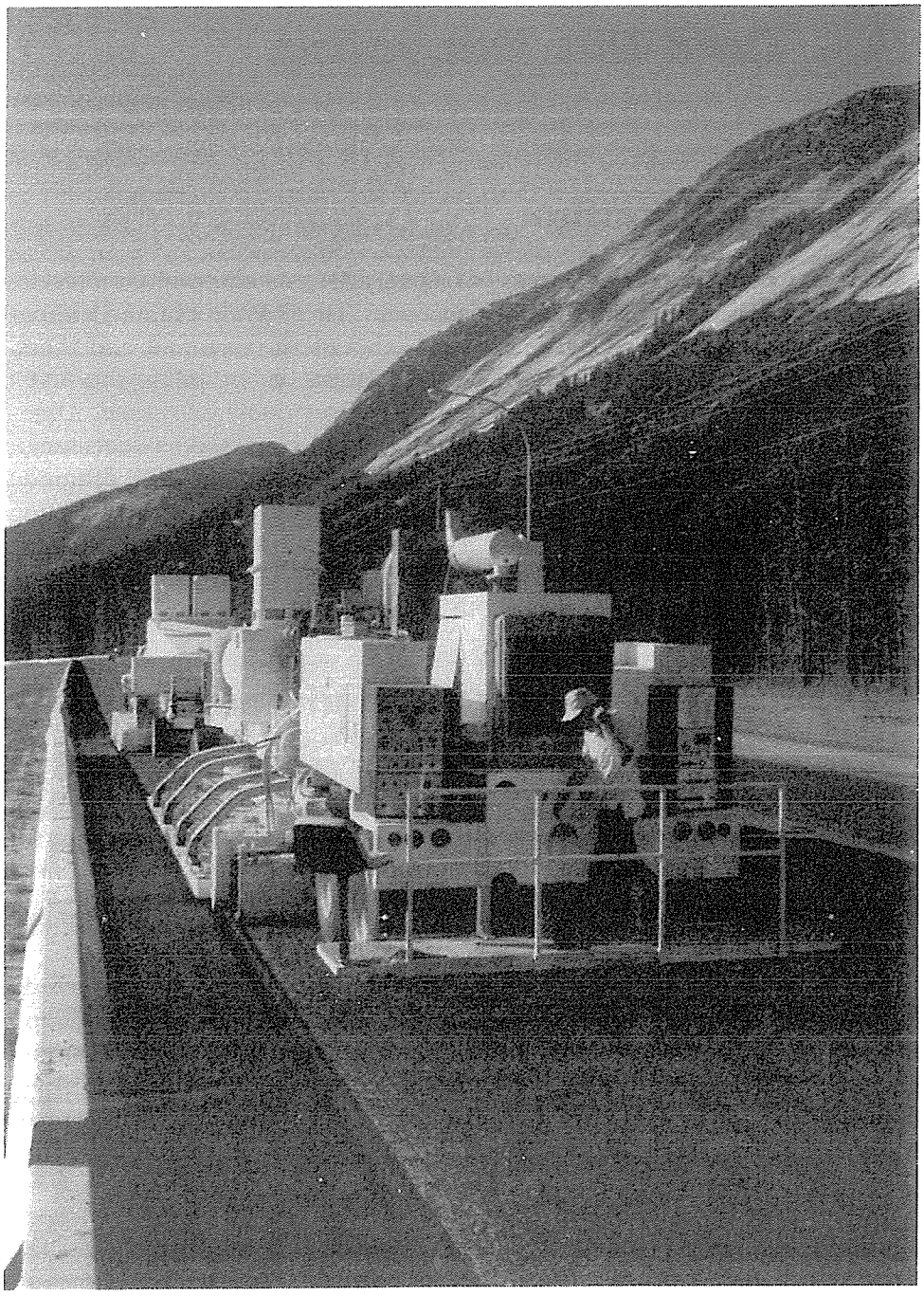


Figure 13.18 Windrow of Material behind First Unit

13.5 Cold-Mix Recycling

13.5.1 Introduction

13.5

Cold-mix recycling is the process of using RAP as a component material when producing a cold mix asphalt material. The RAP is obtained either by excavating the asphalt pavement from an existing pavement and then crushing and screening it to the desired gradation or by milling the RAP from the roadway using milling equipment, as previously discussed. After the RAP has been removed from the roadway and processed as necessary, it is mixed with an asphalt emulsion, a recycling agent, or with a foamed asphalt and additional aggregate (if necessary). The cold-mix recycling process can be accomplished in either a central plant—where the RAP is hauled to a central location similar to how RAP is used as a component in HMA—or in an in-place process on the roadway.

13.5.2 Central Plant (CP) Cold Recycling

General

Millings and recovered asphalt pavement that may need crushing are an excellent source of material for constructing an asphalt-treated base or as a patching material for maintenance of HMA pavements. A discussion of cold mixing is included in Chapter 15, "Special Uses of Asphalt and HMA."

Materials

RAP is produced by milling, ripping, breaking, crushing, or pulverizing the roadway. RAP is hauled to a central location. RAP material handling and storage is similar to that of conventional aggregate. It should be stored in conical stockpiles of 10 feet (3 meters) or less. Conical piles are recommended because RAP has a tendency to re-agglomerate in large piles. A crust will form on the first 8–12 inches (200–250 millimeters) of the pile. The crust and the use of conical piles will help shed water from precipitation. The crust is easily broken with a front-end loader. The RAP stockpile will hold water and not drain like a conventional aggregate stockpile. The use of a horizontal stockpile will result in a greater accumulation of moisture (up to 7–8 percent).

The sources of RAP for central plant cold recycling vary (they can come from different projects, contain chip seals and slurry seals, and/or maintenance patches). Thus, many agencies require that the RAP not be blended or commingled with RAP from other projects. When this policy is followed, the stockpiles should be kept separated and identified by source (preferably on a map and with signs identifying the material in each stockpile). When they are

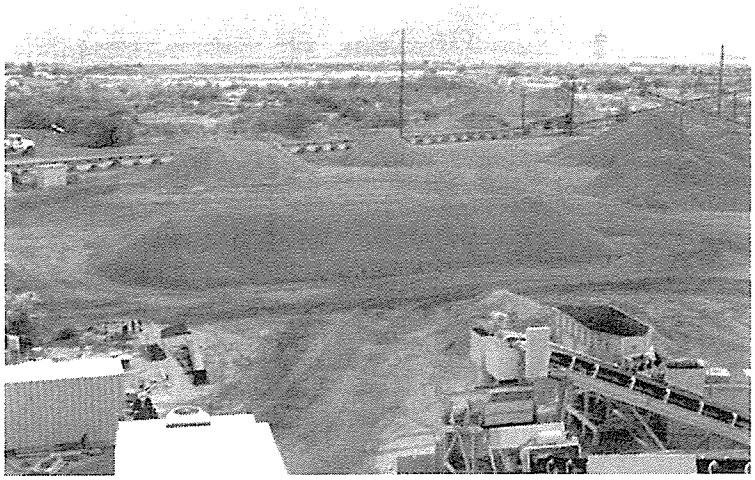


Figure 13.19 RAP Stockpiles

blended for use on a project, the blending process must produce a consistent product. Consistent RAP materials can be produced by crushing and screening the RAP into separate coarse and fine stockpiles. Heavy equipment, such as front-end loaders and bulldozers, should not operate on a stockpile to avoid compacting the material, which would make it difficult to further handle the RAP. See Figure 13.19 for a picture of the proper handling of RAP stockpiles.

Asphalts Used for Cold-Mix Recycling

Various asphalt binders have been used for cold central plant recycling and for cold in-place recycling. Typically cationic, anionic or high-float medium setting (HFMS) emulsions, or polymer-modified versions of these, have been used. A recent development uses a foamed asphalt process in a central mix plant. To increase the initial strength and resistance to water damage, small amounts (1–1½ percent) of either lime or cement can be added to a mix using an asphalt emulsion. They can be added dry or as slurry (lime, cement, and water).

13.5.3 Stationary Facilities

Central plant cold recycling can be accomplished in a standard batch plant or other plant that may be located at a central mixing location, or it can be processed at the job site using a mobile mixing plant. The production of high-quality cold recycled mixes requires attention to detail. The nature of the plant varies according to the size of the project (quantity of mix being produced). The plant should consist of (1) a twin-shaft pugmill equipped with emulsion and water spray bars, (2) trailers for water and emulsion storage, (3) cold feed bins, (4) surge storage, either a stockpile or bins

(needed to ensure a continuous operation of the plant), (5) the necessary conveyor belts, and (6) some sort of mechanical or computer control of the overall process. A more thorough discussion of mix facilities for cold-mix pavements can be found in Chapter 15, "Special Uses of Asphalt and HMA," in the discussion about cold mixes.

13.5.4 Placement of the Mix

The cold recycled mixture can be placed with a standard paving machine or can be spread and shaped with a motor grader. After placement, the mix should be compacted following the same process as that described for cold in-place recycled mixes in the following paragraphs.

13.6 Cold In-Place Recycling

13.6.1 General

The Cold In-Place Recycling (CIPR) process consists of milling the existing pavement to a depth of 3–5 inches (75–125 millimeters), screening, crushing, and mixing with an asphalt emulsion, foamed asphalt, additives (lime or cement), and adding aggregate if needed. The recycled mixture is then placed and compacted using conventional paving equipment or by motor grader.

Project selection is critical to the success or failure of a CIPR project. It must include an assessment of the existing pavement condition, types of distress, and modes of failure. To design the required overlay thickness, the pavement materials (base, subbase, and subgrade) must be sampled and tested. In addition, an evaluation of the history of the pavement maintenance operations and the expected traffic volume must be included in the analysis.

Most pavement distresses, such as fatigue cracking, transverse cracking, reflection cracking, and raveling, can be successfully corrected using CIPR. Using CIPR eliminates damaged layers, resulting in a crack-free layer that provides a base for the HMA surface layer. For CIPR to be effective in providing a long-life pavement structure, as much of the existing HMA layer as possible should be treated. When considering the depth of recycling, the designer must consider the ability of the pavement structure to support the recycling equipment operating on a weakened pavement section as the existing materials are being processed. It is suggested that at least 70 percent of the existing pavement thickness be treated and that at least 1–2 inches (25–50 millimeters) of pavement remain in place to support the weight of the recycling train. If there are paved shoulders, they should be treated to prevent propagation of cracks into the driving lane. See Figure 13.20 for a typical pavement for cold in-place recycling. The right side of the figure shows the cold in-place pavement and the left side is the pavement before recycling.

Badly rutted pavements with excessively high asphalt content can be corrected with CIPR, but additional aggregate may be necessary to offset the high-asphalt content in the existing roadway. Hydrated lime, Portland cement or Class C fly ash are often used as additives to stiffen the mixtures and reduce curing time for CIPR. If additional aggregates are used, they should be crushed angular materials in an amount limited to 25 percent by weight of the RAP. A pavement that failed because of a wet unstable base, subbase, or subgrade is generally not a suitable candidate for CIPR. A pavement exhibiting this type of failure should be considered for full-depth reclamation (discussed in the next section) or for complete reconstruction, along with pavements where the underlying soils are expansive in nature and/or where the asphalt layer is badly stripped.



Figure 13.20 Typical Pavement for Cold In-place Recycling

Other factors that should be considered when evaluating the suitability of using CIPR are these:

- Numerous manholes or drainage outlets
- Steep grades, such as five percent or more
- Asphalt layer thickness less than 2 inches, (50 millimeters) which is not practical for CIPR
- Traffic control on a roadway with a large number of business-access driveways
- Situations where it is infeasible to operate all of the equipment in sequence due to frequent intersections and tight curves.

Cold in-place recycling should not be done in cool, inclement weather. CIPR should be accomplished at temperatures above 50° (10°C), with no freezing weather in the next 24 hours. It should not be done when fog or rain are present or imminent.

13.6.2 Mix Design

There is no nationally recognized method for the design of CIPR mixes, although many agencies have developed procedures for doing so. The simplest methods consist of using empirical equations or the experience of the mix-design engineer to establish suggested asphalt content for the mix based on the amount and consistency of the asphalt binder in the RAP. The complex methods include using resilient modulus, stability, and moisture sensitivity tests. Adjustments are made in the field to the initial

additive (asphalt binder, fly ash, and/or lime) amounts to optimize the performance of the mixture. The following text outlines some of the concerns associated with accomplishing a CIPR mix design. Further information is included in the Asphalt Institute's Manual MS-19, *A Basic Asphalt Emulsion Manual* and in the Asphalt Recycling and Reclaiming Association's *Basic Asphalt Recycling Manual (BARM).*

Sampling Existing Pavement

The first step in obtaining samples of the existing roadway is to conduct a visual survey of the project so that the sampling plan will result in sufficient samples to fully describe the materials in the roadway. It is suggested that samples be taken at a minimum of one per each half mile (kilometer) of roadway. The sample should be full-depth—either a core sample, a bulk sample obtained by sawing or milling, or a combination. A core sample should be at least 6 inches (150 millimeters) in diameter. Samples obtained from coring or sawing can be processed in a laboratory crusher to simulate the effect of a milling machine during construction. The samples used in the mix design should be from the actual layer(s) of the pavement that will be cold recycled. Samples should be obtained during construction to compare with the mix design materials and thus provide a basis for revision of the design quantities for the additives.

Materials Selection and Mix Design Process

The properties of the aggregate and recovered asphalt binder content should be determined. The asphalt binder should be extracted from the RAP using ASTM D 6307 or AASHTO T 308, "Test Method for Asphalt Content of Asphalt Cement of Hot Mix Asphalt by the Ignition Method." After extraction, the asphalt binder content and the gradation of the extracted aggregate should be determined. If additional aggregate needs to be added to improve the mix stability of the final CIPR layer, this is the gradation that should be used. The addition of an aggregate with a high percentage of crushed coarse aggregate can make significant improvements in the stability of the mix. The percentage of new aggregate to be added to the mix should be determined using standard blending procedures to provide a mix with the desired gradation.

Additional asphalt binder is necessary for facilitating compaction, improving the cohesion of the new mix, and reducing the stiffness of the aged asphalt binder in the RAP. Typically, 0.5–1.5 percent additional residual binder is needed. The more uncoated new aggregate used in the mix, the higher the additional binder content will be. For most mix designs, this amount is based on previous experience in ensuring the coating of the RAP material. The amount of binder selected during the mix design for most projects is a starting point, with changes made in the field. The binder can be an emulsified asphalt cationic or anionic slow-set emulsion, or a high-float emulsion. Polymer-modified versions of these materials are also used to improve early strength and to reduce rutting and thermal cracking. The binder can also be a foamed asphalt.

When using emulsions, some moisture generally must be added to provide for coating of the RAP particles. The water can be added at the cutting head of the milling machine or directly to the CIPR mixture. A coating test is recommended. This test consists of mixing a quantity of RAP (typically 2 pounds; 1000 grams) of RAP with various amounts of water at 0.5 percent increments. The water is mixed into the RAP and then the asphalt binder is added. Any admixtures (lime, fly ash, or cement) to be included should be added to the mix before the asphalt binder is added. Adding dust-sized materials will generally increase the amount of moisture needed. They are mixed by hand in a bowl, and the dispersion of the materials is visually monitored. The lowest moisture content that results in no additional increase in curing is selected. The result of the asphalt binder and the added moisture is referred to as the fluids content in the mix.

Mix design samples are manufactured using the RAP, the proposed asphalt binder, and water content (which make up the total fluids content). Generally, the mixing is done at ambient temperatures of 68°–77°F (20°–25°C). The samples are mixed at the total fluid content discussed previously, and the proposed asphalt binder is varied above and below the initial additive content by 0.7 percent. The mix water is adjusted to keep the total fluids content the same for all samples. The samples are either manufactured using a 50-blow Marshall compaction at ambient field conditions or can be compacted with a Superpave gyratory compactor. Studies have shown a compaction level of about 35 gyrations is appropriate.

Before further testing, the samples must be cured to eliminate their excess water and for the emulsions to set up to develop their maximum strength. The curing process varies with the agency doing the testing. The samples are cured in the mold for 4 hours at 140°F (60°C) and then extruded from the mold and cured until they reach a constant mass.

After the samples are manufactured, the volumetric properties are determined using the procedures discussed in Chapter 4, "Hot Mix Asphalt (HMA) Design." CIPR samples may break up during the standard bulk specific gravity test, and thus the CoreLok procedure discussed in Chapter 4 should be used. Strength testing—such as Marshall stability and flow, unconfined compression, and resilient modulus tests—has been used to evaluate the CIPR mixes. But no guidelines have been established. The job-mix formula used as a starting point in the field is based on the coating of the RAP materials and the optimum density of the mix.

13.6.3 Equipment and Methods

Milling and Pulverization

The cold in-place recycling process includes milling the existing pavement to a prescribed depth (typically 2–4 inches (50–100 millimeters) and blending with an asphalt emulsion and additives such as hydrated lime, PCC, or Class C fly ash. Depending on the process, the millings

may be further processed through screening and crushing, and additional aggregate may also be incorporated into the mixture if needed. Processes used include a single-unit, a double-unit, or a multiple-unit train.

Single-Unit Train

The single-unit train consists of a milling machine that does the milling, RAP sizing, and blending at the cutting head. See Figure 13.21. A disadvantage of the single-unit system is that it does not screen and crush the RAP. This makes controlling the maximum particle size very difficult. The RAP size is controlled by the speed of the machine and the speed of the milling head. A grid of 2-inch bars or screen cloth placed over the exit from the milling chamber will act as a scalping device or "grizzly," keeping large pieces in until they are broken down to less than 2 inches. This type of process should not be used for pavements with severe alligator cracking because it is so difficult to control the maximum particle size. A spray bar or foaming nozzles are located in the cutting chamber to add the asphalt emulsion or foamed asphalt. The emulsion is provided either from an internal tank on the milling machine or from a tanker truck pushed or pulled by the milling machine. Aggregate or dry additives to be added to the RAP can be incorporated into the mix by spreading the material on the pavement ahead of the milling operation. The processed RAP is placed in a windrow and picked up by a windrow pickup device, or it is deposited directly into a paver for placement on the roadway.

Two-Unit Trains

The two-unit train consists of a milling machine and a pugmill mixer-paver. The pugmill mixer-paver is a cold-mix paver with an internal pugmill that is used to blend the millings, asphalt emulsion and any additives. The two-unit train does not include any crushing and screening capability. The particle size is controlled in the same manner as in the single-unit train. The RAP is fed directly into a pugmill that has the

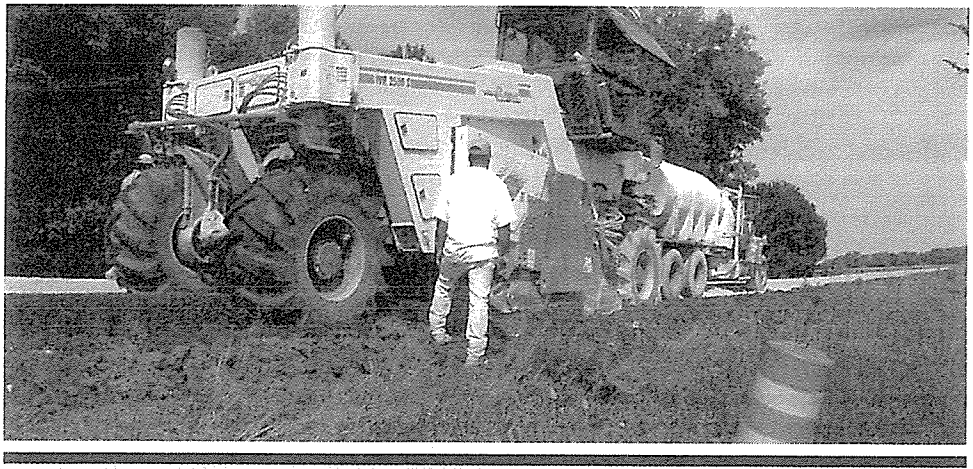


Figure 13.21 Single-Unit CIPR Machine

capability to accurately control the amount of asphalt emulsion and water being added to the RAP. The primary advantage of the two-unit train is its simplicity of operation and high production capability. Two-unit trains are shorter and more maneuverable than multi-unit CIPR trains.

Multi-unit Trains

A multi-unit train consists of a milling machine, a portable screening and crushing unit, and a portable pugmill mixer (see Figure 13.22). The milling machine controls the depth of the pavement being milled. The RAP is picked up by the milling machine and deposited into the screening and crushing unit. The material is passed over a screen deck, and any oversized material is returned to a crushing unit for resizing. After the oversized material has been crushed, it is sent back over the screening unit. The maximum size of the RAP is controlled by the opening size of the bottom two screens. Typically, the maximum particle size is 1–1½ inches (25–37.5 millimeters).

The RAP moves from the processing unit to the pugmill where the asphalt emulsion is added and mixed. The weight of the RAP is monitored as it enters the pugmill. The amount of emulsion is controlled by a computerized metering system. The RAP and emulsion are mixed in a twin-shaft pugmill. The addition of a crushing and screening unit provides better gradation and mixing control than the single and double-unit processes, and it results in a more uniform asphalt-treated base course.

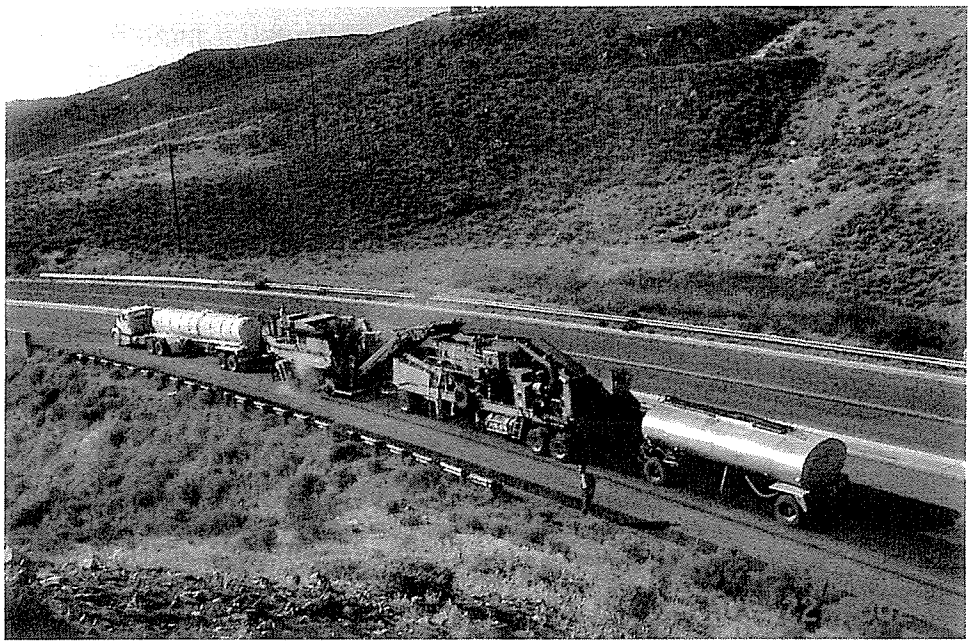


Figure 13.22 Multi-unit CIR Train

13.6.4 Placement of CIPR Materials

The CIPR mix can be placed in a windrow (Figure 13.23) or discharged directly into the paving machine. The procedures discussed in Chapter 8, "Hot Mix Asphalt Paving" should be followed. Conventional paving equipment is used to place the RAP. A paver with a surge bin or hopper insert can be used to reduce the fluctuations in the production of the mix. Unlike HMA paving, the screed must be operated cold because a heated screed causes the asphalt emulsion to break and the mix to stick to the screed. The mix should be handled in such a manner to avoid segregation.

13.6.5 Compaction of CIPR Layers

The compaction of a CIPR mix should be done at a slow and uniform speed. Any change in the direction of the rollers should be done carefully to avoid displacement of the CIPR. Solvents should not be used on the roller wheels or drums to prevent pickup of the CIPR. To prevent adhesion of the CIPR to the rollers, water can be sprayed on the wheels/drums. A small amount of liquid detergent or water-soluble release agent can be added to the water to reduce pickup.

Initial compaction can be done with heavy vibratory, pneumatic rollers, and steel-wheel rollers. It has been found that a CIPR mix cannot be compacted to the same level as HMA. Typically, a CIPR mix will be compacted to an in-place air-void range of 9–14 percent due to the moisture content of the



Figure 13.23 CIPR Mix in a Windrow

mixture. Compaction should begin when the CIPR mix has set enough so that the roller does not cause excessive displacement. The setting of an asphalt emulsion can take from 30 minutes to 2 hours, depending on the ambient conditions. The mix will turn from brown to black when the emulsion breaks. Delaying compaction may result in the formation of a crust on the surface of the mix, making compaction more difficult. Some agencies add lime, Portland cement, or Class C fly ash, to accelerate curing so the roadway can be opened to traffic sooner (caution—type F fly ash should not be used because it does not have pozzolanic characteristics).

Initial compaction should be done with the roller traveling parallel to the pavement's center line. The rolling should begin at each edge and work toward the center. Banked curves should be compacted with the roller starting at the edge of the low side of the bank and then moving toward the superelevated edge. To prevent tearing, gouging, or distortion of the pavement, a minimum of one roller pass should be completed before any turning.

Breakdown rolling should continue until the roller "walks out" of the CIPR mix with either a large pneumatic roller or vibratory steel-wheel roller operating in static mode. Some agencies do not recommend a steel-wheel roller for breakdown rolling because of a concern that sealing moisture in the mix will delay curing. Breakdown rolling is followed by intermediate rolling with a 12-ton (10,800 kilogram) or more double-drum vibratory roller in static mode to remove the roller marks (see Figure 13.24).

Rolling patterns should be established using a nuclear density gauge on a designated control or test strip. Different combinations with different rollers should be evaluated. When the results of a rolling pattern do

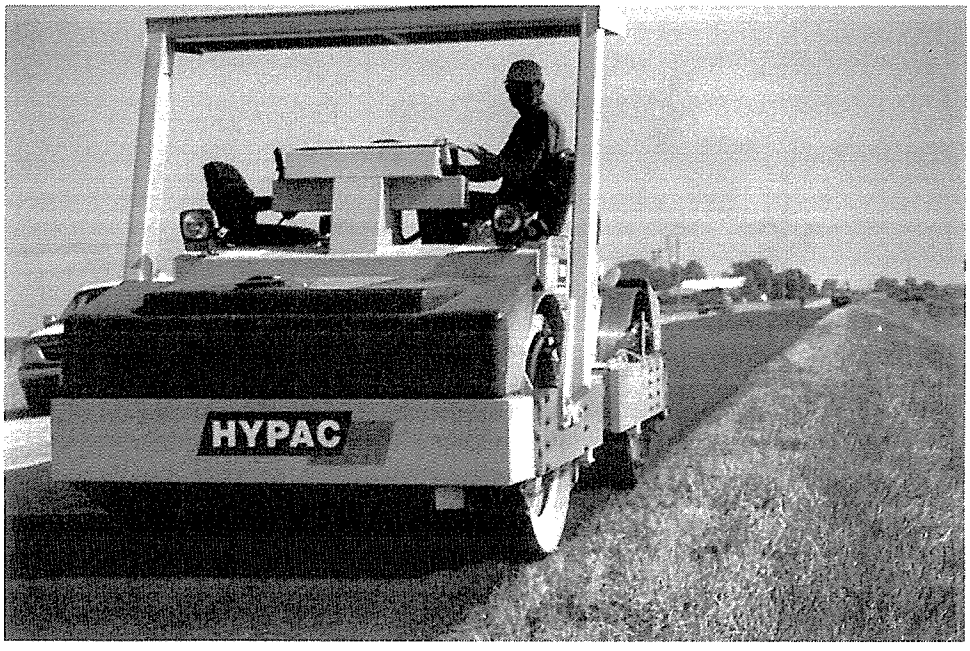


Figure 13.24 Steel-Wheel Finish Roller

not show any increase in the density of the mix, that pattern should be selected. Once the test strip density is established, the remaining project should be compacted to a minimum of 97 percent of the relative density of the control strip. Monitoring the density should continue, and if it is found that density can be obtained too easily or with great difficulty, a change in the mixture and a new roller pattern and/or new control density might be needed.

After compacting, the pavement can be opened to traffic. Before opening the road to traffic, the pavement and the shoulders should be swept to remove loose rock or CIPR.

13.6.6 Curing and Aeration

When using emulsions, the compacted CIPR mix must be cured before a wearing course can be placed. Curing is needed to reduce the moisture content in the mixture before it is sealed by the wearing/surface courses. The time required for this process depends on daytime and nighttime temperatures, the humidity levels and rainfall after completion of the recycling process, and the amount of water added to the mix during the CIPR process. If necessary, laydown and compaction can be delayed and the mix aerated with a motorgrader to accelerate the curing process. If rain is forecasted the CIPR mix can be bladed flat and sealed with a steel wheeled roller until the weather clears up and aeration and curing can continue.

Different agencies define this moisture level on the basis of total moisture in the mix or on the increase in the moisture content in the pavement before recycling. Some agencies feel that if a dry core is obtained from CIPR mix relatively intact, the mix has cured enough to place it on the wearing surface. This curing/aeration process can take 10 to 14 days. In advance, if this timeframe is a problem, hydrated lime or Portland cement can be added during the mixing process to accelerate curing.

During the curing process, traffic is generally allowed to operate on the CIPR surface. In fact, the kneading action of traffic can aid the aeration process. If minor rutting occurs during the curing aeration period, the surface of the CIPR mix can be lightly scarified, bladed and re-rolled using a heavy pneumatic roller following by a static steel roller to remove any roller marks. This rolling should be done on a hot day during the heat of the day. Another problem that can occur is raveling of the surface aggregate on the CIPR. If this occurs, a fog seal of diluted emulsion (50 percent with water) can be applied at a rate of 0.05–0.10 gallon/yard^2 (0.23–0.45 litre/meter^2). The application of a fog seal should not be done unless absolutely necessary and traffic can be kept off the surface until the fog seal has had a chance to completely dry (at least 2 hours) because the fog seal can seal the CIPR surface and impede the curing process. When foamed asphalt is used for CIPR, no curing or aeration is required. The small amount of water used in the foaming process flashes to steam when it comes in contact with the hot liquid asphalt leaving no moisture behind.

13.6.7 Wearing Course

CIPR mixes are not intended to be a finished surface or wearing course and will require a wearing course because of the high air-voids and to protect the surface from raveling. For pavements with low traffic volumes, single- and double-surface treatments have been used. For pavements with a higher traffic volume, the thickness-design procedures detailed in Chapter 6, "Thickness Design" should be used.

13.6.8 Inspection

Roadway Inspection

During the construction process, the following should be checked:

- The weather and seasonal requirements in the specifications must be met.
- If aggregate and/or additive is being added to the mix by spreading in front of the milling machine, periodic yield checks should be made.
- The yield of the emulsion, aggregate and additive should be computed and compared to the mix design.
- The CIPR mix should be visually checked to ensure that it is homogeneous, with no segregation; it should also be checked to ensure that no large (sometimes agglomerated particles or uncrushed pieces) are in the mix. If this occurs, the crushing and screening process should be checked.
- The coating of the mix should be checked. If the mix is not adequately coated, the amount of mix water should be increased. Excessive water can cause the asphalt to flush to the surface and increase the curing time. Too little water may cause segregation. Reducing the asphalt emulsion one grade (for example, using an HFE 300 instead of an HFE 150) may also improve coating.
- The CIPR mix should be checked for cohesion. If the mix lacks cohesion, the emulsion content should be increased. Too much asphalt emulsion can result in an unstable mix; too little can cause the mix to ravel. A simple field test can be used to evaluate cohesion. A ball of the material is made by squeezing the material in the fist. If the ball falls apart after the pressure is released, the mix lacks cohesion. The palm of the hand should contain specks of asphalt, indicating proper asphalt content. If the hand is stained, the asphalt content is too high.
- Over-mixing can lead to scrubbing the emulsion off of the aggregate particles and early emulsion break that causes the mix to be too stiff. Under-mixing results in undercoating of the aggregate. 100 percent coating, as in HMA, is not the goal with CIPR mixes.
- The binder content can also be evaluated during placement of the CIPR mix after initial compaction. The mat should be brown and cohesive. Excessive raveling indicates too little asphalt in the mix.

A shiny black surface indicates too much asphalt. Adjustments should be made by experienced personnel in 0.2 percent increments of emulsion content.

- The inspector should be aware of any large maintenance patches in the existing roadway. It may be necessary to remove those sections prior to CIPR and replace them with either HMA or RAP. Maintenance patches are usually manufactured and placed by maintenance forces using cutback asphalts. To improve workability, excess asphalt is sometimes used in these patches. If such patches are left in the roadway and not removed, it may be necessary to adjust the amount of asphalt being added to the process as the CIPR machine moves through the patch.
- Quality-control testing should consist of taking random samples of the CIPR in accordance with the agency's guidelines—or at least two samples per mile (one per kilometer). These samples should be tested for aggregate gradation and asphalt content.
- The ride quality should be checked daily by driving through the project. Ride specifications are not generally incorporated into CIPR projects. The steps discussed in Chapter 8. "Hot Mix Asphalt Paving"— for insuring a smooth pavement surface—also apply to CIPR projects.

13.7 Full-Depth Reclamation

Full-depth Reclamation (FDR) consists of pulverizing the full thickness of the asphalt pavement and a portion of the underlying base, subbase, and/or subgrade. The layers are blended to form a homogeneous, stabilized base layer. This pulverized layer can be stabilized mechanically, chemically, asphalt stabilized or a combination of the approaches. The *Basic Asphalt Recycling Manual* prepared by the Asphalt Recycling and Reclaiming Association is an excellent resource on the FDR process.

Mechanical stabilization, which may be accomplished by adding crushed aggregate, RAP, or crushed PCC, increases the strength solely through improving the aggregate interlock. Chemical stabilization uses lime, cement, or lime/fly ash to form a binder, while asphalt stabilization uses asphalt emulsions or foamed asphalt.

The type, severity, and frequency of pavement distress in the existing pavement should be evaluated during the project's design process. The FDR process destroys the existing pavement to produce a homogenous pulverized layer. If the existing pavement has large or frequent patching, the variability of the final product will be affected. In areas where the patches are the result of subgrade instability or in high water-table areas, the cause of the problem should be fixed before beginning the FDR process. This may require excavating the problem areas and replacing some subgrade with improved materials, or improving the drainage system by installing a subdrainage system or reworking the ditches in the area.

The pavement structure should be designed according to the procedures detailed in Chapter 6, "Thickness Design," using the appropriate design values for the FDR layers.

13.7.1 Stabilization with Chemical Additives

When chemical stabilization is being evaluated for use in FDR, the following criteria should be used to evaluate which chemical should be selected:

- **Lime:** If the PI $>$ 20 and the minimum clay content (2 microns) is $>$ 10 percent
- **Cement:** If the PI $\leq$ 10 percent and the percent passing the No. 200 (.075 millimeter) sieve is $<$ 20 percent
- **Lime, cement, or fly ash combination:** If layer has material with less than 10 percent passing the No. 200 (0.075 millimeter) sieve and the Plasticity Index range is 10$<$PI$<$20